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Project Report

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Project Name

Supervised Classification and Anomaly Detection of Transient Supernovae in ZTF and LSST Alert Streams: An End-to-End Pipeline with Measured Sim-to-Real Transfer

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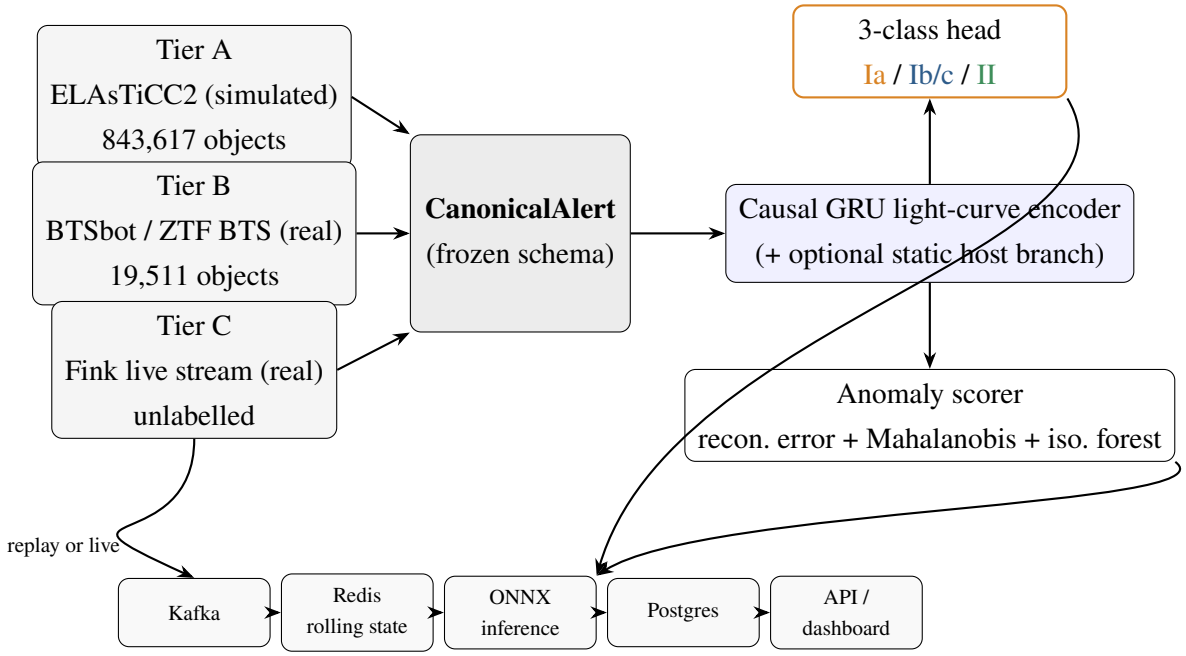
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Abstract

We describe an end-to-end pipeline classifying transient alerts as Type Ia, Ib/c, or II supernovae, or flagging those resembling none, across three data tiers: simulated ELAsTiCC2 light curves (843617 objects, Tier A) for pretraining, real labelled ZTF light curves (BTSbot, Tier B) for fine-tuning, and a live Fink stream (ZTF and LSST, Tier C) for serving. The classifier is a causal gated recurrent unit over per-observation flux, conditioned on continuous filter wavelength rather than band identity, so weights learned on simulated LSST photometry transfer to real ZTF photometry. A separate branch scores anomalies via light-curve reconstruction error, Mahalanobis distance in the classifier's latent space, and an isolation forest. On held-out Tier A data the classifier reaches macro-F1 0.7169 (Ia-vs-rest ROC-AUC 0.8822); fine-tuned on Tier B, the deployed model reaches 0.6716 (a 0.0454 drop, the expected sim-to-real direction), and fine-tuning recovers 0.1860 macro-F1 over the zero-shot baseline. The anomaly branch's three-component combination reaches ROC-AUC 0.8146 on held-out simulated exotic classes and 0.7622 on real confirmed non-supernova objects – smaller but useful under domain shift. The pipeline was validated against four pre-registered gates (reader correctness, causality/label-shuffling, sim-to-real drop, manual anomaly inspection) and briefly run against live ZTF and LSST alert traffic; a serving latency benchmark exists in project records but is not restated here since no artifact survives. We report two unanticipated results – a net-negative effect of host-galaxy features on real-data fine-tuning, and an anomaly classifier that grows more, not less, confident on unfamiliar inputs – as open findings.

Project Visual



Pipeline architecture. Three data tiers (simulated ELAsTiCC2 for pretraining, real ZTF/BTSbot for fine-tuning, live Fink for serving) convert into one canonical alert schema, which every downstream stage consumes identically regardless of source. A shared causal GRU light-curve encoder (optionally fused with a static host-context branch) feeds both the flat 3-class classification head and a separate anomaly-scoring branch. The serving path (Kafka → Redis rolling state → ONNX inference → Postgres → API) is the same consumer code whether its producer is an archived-data replay or the live Fink stream. This same diagram recurs as Figure 3.1 in Chapter 3, where each stage is discussed in detail.

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Chapter 1

Introduction

A supernova is the catastrophic, once-per-object brightening of a star (or stellar remnant) at the end point of its evolution. For a period of days to months it can rival or exceed the integrated brightness of its entire host galaxy, which is what makes supernovae visible across cosmological distances and useful as tracers of the universe's structure and expansion. Astronomers group observed supernovae by *spectroscopic type*: what absorption and emission features appear when the transient's light is dispersed through a spectrograph. This report is concerned with three of those types, and with a fourth category – "none of the above" – that a real alert stream inevitably also produces.

Spectroscopy is definitive; photometry is what scales. A spectrum directly measures the physics – the elements present, their ionisation state, the ejecta's expansion velocity – and is the only fully reliable way to assign a supernova's type. It is also expensive: it requires a sufficiently large telescope, a spectrograph, and dedicated observing time allocated to one object at a time. What a wide-field synoptic survey produces automatically, for every object, every time it is observed, is a *light curve*: a sequence of calibrated flux measurements at a handful of timestamps and wavelength filters ("bands"). The central question this project asks is how much of the spectroscopic answer – Ia, Ib/c, or II – can be recovered from a light curve's shape alone: how fast it rises, how it declines, whether it plateaus, all without ever taking a spectrum. Doing this *early*, before the light curve has finished playing out, is the useful case in practice: a system that can say "point a spectrograph at this one" while the transient is still spectroscopically interesting is worth far more than one that can only classify a supernova after it has faded.

The scale problem. The Zwicky Transient Facility (ZTF) already produces alert volumes that strain manual and even semi-automated triage (Bellm et al., 2019). The Vera C. Rubin Observatory's Legacy Survey of Space and Time (LSST), whose main ten-year survey began operations on 30 June 2026, is designed to generate on the order of 10^7 transient and variable-source alerts

per night at full operational cadence (Ivezić et al., 2019) – a volume no spectroscopic follow-up programme, however well resourced, can classify object by object. Automated, photometry-only triage is therefore not an optional convenience for the LSST era; it is the only mechanism by which a human-in-the-loop follow-up programme can be pointed at the right handful of objects out of millions. This report describes one such triage system, built and measured against both simulated LSST-like data and real ZTF data, and served, briefly, over a live alert stream from both surveys.

1.1 Motive and Objectives

The programme brief, as set by the Summer School of Astronomy and Astrophysics, asked for a multi-survey, multimodal pipeline that classifies transient alerts into supernova subtypes and flags objects that fit none of them, pretrained on simulated data, fine-tuned on real spectroscopically labelled data, served over a live alert stream, and refined by a human-in-the-loop anomaly review process. From that brief, seven concrete, individually testable objectives were set at the start of the project:

1. Build a single canonical data schema and one reader per data source (simulated ELAsTiCC2, real ZTF/BTSbot, live Fink) that converts each source into it, so that a model trained on one source can be evaluated on another without a source-specific code path.

2. Train a causal light-curve classifier on simulated data alone and establish its ceiling performance and failure modes before any real data is introduced.

3. Fine-tune that classifier on real, spectroscopically labelled data and *measure* the resulting sim-to-real accuracy gap, rather than assume fine-tuning succeeds.

4. Build a separate anomaly-detection branch capable of ranking alerts that do not resemble any of the three target classes, and evaluate it against both a simulated held-out anomaly population and a real one.

5. Serve the trained models over a Kafka-based pipeline, first in replay mode against archived data and then, unmodified, against a live alert stream from at least one real broker.

6. Validate every stage against pre-registered acceptance criteria (Section 3.1's four gates) defined *before* the results they judge, rather than criteria chosen after the fact to match whatever result was obtained.

7. Produce a results and monitoring interface that makes every reported number traceable to the artifact that produced it.

Sections 3.2–4 report the outcome of each objective in turn; Section 5 discusses which were fully met, which were only partially met, and why.

Chapter 2

Theoretical Framework and Literature Review

2.1 Theoretical Framework

Two unrelated physical channels, one shared naming scheme. Type Ia supernovae arise from the thermonuclear disruption of a carbon-oxygen white dwarf, either through mass accretion from a companion star past a critical threshold or through the merger of two white dwarfs. The explosion consumes the star with no hydrogen involved at any stage, which is why a Ia spectrum shows no hydrogen lines. Type Ib/c and Type II supernovae, by contrast, are both *core-collapse* events: a massive star ($\gtrsim 8\,M_{\odot}$) exhausts its nuclear fuel, its iron core can no longer support itself against gravity, and the resulting collapse (typically to a neutron star) drives a shock that unbinds the star's outer layers. Type II supernovae still have their hydrogen envelope at the moment of collapse; Type Ib and Ic supernovae have already lost it (and, for Ic, their helium layer too) to strong stellar winds or to a binary companion. It is worth stating plainly, because the shared Roman-numeral naming scheme actively obscures it: *Type Ia is not physically related to Type Ib or Ic*. The "I" in each name refers only to the historical absence of hydrogen lines in the discovery-era spectroscopic classification scheme (Filippenko, 1997), not to a shared progenitor or explosion mechanism. Type Ib and Ic are two flavours of the same core-collapse channel as Type II; Type Ia is a different channel entirely. This project follows convention (and its own taxonomy, Section 3.2) in merging Ib and Ic into one target class, since distinguishing a stripped-envelope supernova that has lost its hydrogen from one that has lost its helium as well requires spectroscopy this pipeline does not have.

Why Type Ia matters beyond stellar physics. Because a Type Ia explosion detonates a white dwarf at (or close to) a common physical trigger mass, the resulting explosions are photometrically similar to one another: their peak absolute brightness is *standardisable* through empirical relations

between peak luminosity, light-curve decline rate, and colour. That standardisability is what turns a Type Ia supernova into a "standard candle" – an object whose true brightness can be inferred well enough from its light curve alone that comparing it to its *observed* brightness yields a distance. Type Ia supernovae were the tool used to discover the accelerating expansion of the universe, and they remain a primary distance-ladder rung today. Recovering Ia identity from photometry alone, quickly, is therefore not merely a pattern-recognition exercise: it is a prerequisite for using a given object cosmologically at all.

2.2 Literature Review

Recurrent networks on raw light curves. Charnock and Moss (2017) were among the first to show that a recurrent neural network fed raw, un-engineered light-curve photometry directly could classify supernova types competitively with feature-engineering pipelines, on simulated Dark Energy Survey-like data – establishing that a sequence model need not first be handed hand-built summary statistics (rise time, colour at peak, and so on) to find useful structure in a light curve. Muthukrishna et al. (2019) extended this to *early*, time-resolved classification with RAPID, a Gated Recurrent Unit (GRU) architecture that emits a classification at every observed timestep rather than only at the end of a light curve, across 12 transient classes; this per-timestep prediction convention, and the resulting practice of reporting accuracy as a function of time since first detection rather than as one end-of-light- curve number, is used throughout the present work (Section 3.3). Möller and de Boissière (2020) (SuperNNova) placed a Bayesian recurrent architecture in this same family, adding calibrated uncertainty estimates and a systematic comparison against feature-based random-forest baselines, run at ZTF and LSST alert-stream scale.

Data challenges. Kessler et al. (2019) designed and described PLAsTiCC, the Photometric LSST Astronomical Time-series Classification Challenge: a large, publicly released simulated dataset spanning many transient and variable-source classes, intended to let classification methods be compared on a common, labelled corpus before real LSST data existed. ELAsTiCC2 (LSST Dark Energy Science Collaboration, 2023), the LSST Dark Energy Science Collaboration's Extended LSST Astronomical Time-series Classification Challenge and this project's own Tier A simulated corpus (Section 3.2), supersedes PLAsTiCC with an updated simulation pipeline, revised class definitions, and – the property this project depends on most directly – deliberately *non*-representative class proportions relative to the real sky rate, so that a classifier trained on it must be evaluated with that mismatch in mind rather than assumed to inherit a realistic prior (Section 3.2).

Current state of the art. Cabrera-Vives et al. (2024) report ATAT, a pair of transformer encoders (one over the light curve, one over tabular host/metadata features) fused before a classification head, reaching a macro-F1 of 82.9 ± 0.4 across 20 ELAsTiCC classes – the strongest published number on this class of problem to date, and the anchor this project's own, considerably smaller 3-class macro-F1 figures should be read against, not compared to directly (Section 4). Shah et al. (2025) (ORACLE) instead build a GRU-based hierarchical classifier trained with a hierarchical cross-entropy loss and host-galaxy context, explicitly targeting the same graceful-degradation property (a correct "supernova-like" call even when the fine-grained subtype call is wrong) that this project's own hierarchical loss experiment tests and does not, in the end, adopt (Section 3.1). A causal-transformer architecture for this same task is described in Tung (2025), reporting 71.8% light-curve-only and 89.8% light-curve-plus-metadata F1 on their own evaluation. Fine-grained supernova subtyping is consistently reported, across this literature, as a harder discrimination axis than the coarser transient-vs-non-transient call – consistent with this project's own recorded expectation that the Ib/c class in particular should be, and is, the hardest of the three target classes to recover cleanly.

Anomaly detection over alert streams. Malanchev et al. (2021) ran an unsupervised anomaly detector (isolation-forest- based) over ZTF's third public data release and reported that of the highest-ranked, most anomalous objects a human inspected, 68% were artifacts of imperfect image subtraction rather than genuine unusual transients – the published anchor this project's own Gate D (deferred, not passed – Section 4) was designed to be checked against. Villar et al. (2021) (VRAENN) instead trained a variational recurrent autoencoder for the same live-anomaly-detection task, an architecture whose reconstruction-error principle is close in spirit to the decoder component of this project's own three-part anomaly score (Section 3.1). `coniferest` (Kornilov et al., 2024), the isolation-forest and active-learning package used here for one of the three anomaly components, is developed and maintained by the SNAD (SNAD Anomaly Detection) team; the production Fink broker's own anomaly-ranking module, described in Möller et al. (2021), is built on the same package.

Real data and brokers. The Zwicky Transient Facility Bright Transient Survey (BTS, Fremling et al., 2020; Perley et al., 2020) is the source of this project's real, spectroscopically labelled fine-tuning data (Tier B). BTSbot (Rehemtulla et al., 2024) is a real-bogus and bright-transient-vs-reject triage classifier trained on the same BTS alert stream; this project reuses its public metadata release as the raw Tier B corpus (Section 3.2) but not its own triage labels, which answer a different question ("is this a real, bright transient at all") from the one this project trains on ("which of Ia/Ib-c/II is it"). ALeRCE (Förster et al., 2021) and Fink (Möller et al., 2021) are both production alert brokers that ingest ZTF (and, for Fink, LSST) alert streams and run their own classification and anomaly-

ranking pipelines in real time; Fink is the broker this project's own serving layer (Section 3.1) actually connects to for live Tier C data.

Transfer learning. Gupta et al. (2025) report that pretraining a transient classifier on simulated data and fine-tuning on a comparatively small real labelled ZTF set reduces the amount of real labelled data needed by about 75% relative to training on real data alone, and that a model pretrained on ZTF reaches 94% of a from-scratch baseline's performance on simulated LSST data using only 30% as much LSST training data – the same sim-to-real transfer strategy this project's own Tier A → Tier B pipeline follows (in the opposite direction, LSST-like simulation to real ZTF), and the same comparison (zero-shot Tier-A-only performance versus fine-tuned performance on Tier B) this project measures directly in Section 4.

Loss design. Villar et al. (2023) propose weighted hierarchical cross-entropy (WHXE): a loss that predicts at every level of a class taxonomy tree, not only at the leaf, and weights a mistake by how far apart the true and predicted classes sit in that tree – confusing two supernova subtypes should cost less than confusing a supernova with a non-supernova. This project implements WHXE essentially as published, encounters and resolves a real ambiguity in the paper's own depth-indexing convention, and reports a controlled, one-run comparison against a flat classification head that does not favour WHXE on this project's own single-branch dataset (Section 3.1); this is stated as a property of the data available at the time of the comparison, not as a criticism of the loss itself.

Surveys. Bellm et al. (2019) describe ZTF's system design, cadence, and early performance; Ivezić et al. (2019) describe LSST's science drivers, reference design, and anticipated data products, including the alert-volume figures cited in Section 1.

Taxonomy. Filippenko (1997) is the standard review of the spectroscopic classification scheme this entire field, and this project's own taxonomy (Section 3.2), is built on.

Where this work sits. None of the architectural components used here are novel: a causal GRU light-curve encoder, a reconstruction-error/Mahalanobis/isolation-forest anomaly ensemble, and a Kafka-based serving layer are all established techniques, individually published elsewhere in the works cited above. What this report adds is not a new architecture but a measured, audited account of what happens when a comparatively standard pipeline of this kind is actually built end to end on a strict timeline: a quantified sim-to-real transfer gap and transfer gain (Section 4); a documented account of where the train-time and serve-time data distributions silently diverge and what that costs (Section 3.3); a measured, non-obvious failure mode in the anomaly ensemble under domain

shift (Section 4); and end-to-end serving and live-ingestion results, run against real ZTF and LSST alert traffic, that are not published elsewhere for a pipeline of this specific shape. Understating this contribution is the safer error to make, and is made deliberately.

Chapter 3

Methodology, Data & Observations

3.1 Methodology

This section states the design principles adopted *before* any model was trained or any result existed, the resulting architecture, and the validation protocol used to judge every result reported in Section 4. Table 3.4 at the end of this section is a decision log: for each major design choice, the alternatives considered, the choice made, and the evidence that settled it. Most write-ups of a pipeline like this present only the final configuration; the alternatives that were rejected, and why, are at least as informative.

3.1.1 Design principles

Wavelength conditioning, not band identity. ZTF alerts carry three photometric filters; LSST alerts carry six (Table 3.1). If a model's input encoded "filter index 2," ZTF's second filter and LSST's second filter would be unrelated integers occupying unrelated positions in the model's input space, and nothing learned from simulated LSST photometry (Tier A) would transfer to real ZTF photometry (Tier B) at all. Instead, every observation in this pipeline carries its filter's actual *effective wavelength*, a continuous physical quantity, linearly mapped into $[-1, 1]$ (Section 3.3). ZTF's r band (633.9 nm) and LSST's r band (622.2 nm) land near each other in the model's input space automatically, because they are physically similar wavelengths, not because any code special-cases "these are both called r." This single decision is what makes "pretrain on simulated LSST photometry, fine-tune on real ZTF photometry" a coherent strategy rather than two unrelated models sharing a name.

Causal models only. Every recurrent layer in this pipeline runs in a single direction only (`bidirectional=False` throughout), with no attention mechanism that could attend to a future

Table 3.1: Filter effective wavelengths used for wavelength conditioning (Section 3.1). LSST's six bands and ZTF's three are mapped into the same continuous space; no band index is ever used as a model input.

LSST		ZTF	
Band	λ_{eff} (nm)	Band	λ_{eff} (nm)
u	368.6	g	472.3
g	480.3	r	633.9
r	622.2	i	788.6
i	754.5		
z	869.1		
y	971.0		

timestep. A bidirectional model evaluated offline, where every object's complete light curve already exists in the training set, would silently learn to use information from an object's future relative to whatever point it is nominally classifying at – on day 3 of a real alert stream, day 60 has not happened yet. Such a model scores well offline and is meaningless in production, and the failure would not be visible in any ordinary train/test split, since both sides of that split share the same non-causal leakage. This is treated as an absolute architectural constraint, not a tunable hyperparameter, and is enforced by a dedicated unit test that truncates a real input sequence at a timestep t and asserts the model's prediction at t is bit-identical whether or not timesteps after t are present in the input tensor at all (Table 3.3, Gate B).

A frozen evaluation harness. The canonical data schema and the evaluation metrics code were written and frozen before any model existed, and were never modified afterward except by one documented, additive amendment (below). The reasoning is that a metric implementation that can still be edited is a metric that will, eventually, be adjusted – consciously or not – to make a disappointing result look better, and the resulting number would be unfalsifiable: there would be no way to tell "the model improved" from "the yardstick moved" after the fact. The one amendment made to the frozen harness after Day 1 was the addition of ROC-AUC (Ia-vs-rest) as a new metric, once the project's own literature anchors (Section 2.2) turned out to name a metric that did not yet exist anywhere in the harness. The amendment added a new function and the minimal wiring to carry its output through the existing report structures, touched no existing function's computed output, and was verified independently (a new value on an existing checkpoint, not a retroactive recomputation of an existing reported number). An amendment of this kind – additive, independently verified, motivated by a real gap rather than a disappointing result – is judged preferable to a blanket no-changes rule that would otherwise have blocked a metric the project's own anchors required.

Flux, never magnitude. Difference-image photometry – subtracting a reference template from a new science exposure – routinely produces negative flux values on noisy, faint, non-detections. Those points are concentrated in exactly the early, pre-peak regime this project's early-classification goal cares about most. A magnitude is a logarithm of flux and cannot represent a negative value at all; any reader that converts to magnitude before this pipeline sees the data has already silently discarded some of the most informative early-time points. Every reader in this pipeline therefore emits flux, on one shared zeropoint (Section 3.2), and only ever converts to magnitude for plotting.

A canonical alert schema as the single conversion target. Every data source – simulated ELAsTiCC2, real ZTF/BTSbot, live Fink – is converted into one shared `CanonicalAlert` record before any model or evaluation code ever touches it (Table 3.2). This is what makes "pretrain on one source, fine-tune on another, serve a third" possible without a source-specific branch anywhere downstream of the reader layer.

Table 3.2: The canonical alert schema (`src/schema.py`, frozen after Day 1–2 of the project). Every reader converts its source into this record; no downstream code depends on which survey an alert came from.

Field	Meaning
`object_id`	Survey-native identifier
`survey`	ZTF / LSST / ELAsTiCC
`observations`	List of (mjd, λ_{eff}, flux, flux_err, is_detection), sorted ascending by mjd
`static`	Host photo-z, offset, ellipticity, host magnitude, MW $E(B-V)$, galactic latitude, real-bogus score
`cutouts`	$(3, 63, 63)$ science/reference/difference postage stamps, or `None`
`class_id`	ELAsTiCC2 taxonomy id (Table 3.7), or `None` if unlabelled

3.1.2 Figure 1: pipeline architecture

Figure 3.1 is the anchor figure for this report: the three data tiers on the left, the canonical schema as the single conversion funnel every tier passes through, the shared light-curve encoder feeding both the classification head and the anomaly scorer, and the serving path from a live alert stream through to a results interface.

3.1.3 Model architecture

The light-curve encoder (Figure 3.1) is a 2-layer causal GRU, hidden size 128, with a `LayerNorm` and dropout ($p = 0.2$) applied to the GRU's output before a linear head. Sequences are packed and

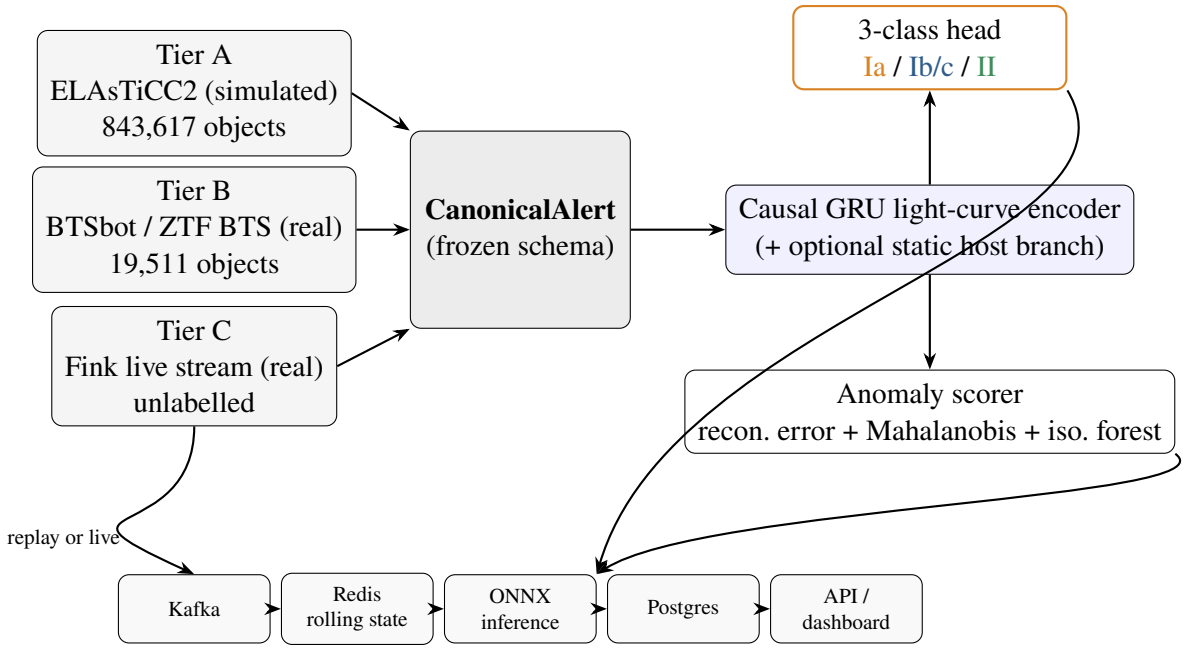


Figure 3.1: Pipeline architecture. Three data tiers (simulated ELAsTiCC2 for pretraining, real ZTF/BTSbot for fine-tuning, live Fink for serving) convert into one canonical alert schema, which every downstream stage consumes identically regardless of source. A shared causal GRU light-curve encoder (optionally fused with a static host-context branch) feeds both the flat 3-class classification head and a separate anomaly-scoring branch. The serving path (Kafka → Redis rolling state → ONNX inference → Postgres → API) is the same consumer code whether its producer is an archived-data replay or the live Fink stream.

masked (`pack_padded_sequence`) so that a padded timestep is never processed by the recurrent layer for any sequence in a batch, regardless of where it sits in the batch. Predictions are emitted at *every* timestep, not only the last, which is what makes the phase-cutoff curves in Section 4 possible without a separate model per cutoff.

Each observation is represented by six per-timestep features: Δt (days since the object's own first detection), signed_log1p(flux/flux_err) (a compressed signal-to-noise ratio), signed_log1p(flux) (compressed absolute flux), log $1p$(flux_err), the linearly-mapped filter wavelength, and a detection/forced-photometry indicator. Absolute flux is carried *in addition to* the signal-to-noise ratio, not instead of it, even though it is in principle recoverable as their product (flux = (flux/flux_err) × flux_err): a Type Ia supernova's absolute peak brightness clusters tightly near $M \approx -19.3$, while Ib/c and II are both fainter and more scattered, so absolute brightness is real discriminating signal for exactly the Ia/non-Ia boundary this pipeline is most sensitive on – but that signal is only accessible as a *multiplicative* interaction between two other, already nonlinearly-compressed input channels, and GRUs and MLPs are comparatively poor at learning multiplicative interactions between their own inputs. Supplying signed_log1p(flux) directly removes that burden

at no additional leakage cost (Section 3.3).

An optional static-context branch (a small MLP over seven host/extinction fields) is fused with the GRU's output before the head in the "LC+static" configuration (Section 4). Missingness in these fields is handled explicitly: each field carries its own zero-filled, z-scored value *and* a separate missingness indicator, rather than a silently imputed value alone. A zero-fill on already z-scored data lands at the training-set mean, a neutral placeholder in the untransformed scale as well; the indicator lets the downstream MLP learn to treat a filled value differently from a measured one, which a silent imputation would not allow it to do at all.

3.1.4 Anomaly branch

The anomaly score combines three components, computed over the encoder's own latent representation and the observed light curve:

- **Reconstruction error**: a small decoder predicts each timestep's flux-derived features from the encoder's hidden state as of the *previous* timestep only (never the current one, which the GRU has already consumed and would make reconstruction trivial), trained on the three normal classes only. An object whose flux behaviour this decoder cannot predict well is anomalous by construction.
- **Mahalanobis distance** of an object's latent encoding from a Gaussian fitted to the normal-class population's latent space.
- **Isolation forest** (`coniferest`'s `PineForest`, used here in pure unsupervised mode) over the concatenation of the latent encoding and a handful of directly-engineered light-curve summary statistics.

The three components are z-scored against their own fit-population statistics and combined by an equal-weighted average (Section 4 discusses why entropy, a fourth candidate component, was tried and dropped).

3.1.5 Hierarchical loss (WHXE)

Weighted hierarchical cross-entropy (Villar et al., 2023) was implemented and compared, in one controlled run (same encoder, data, split, seed, and class weighting as the flat-head baseline), against the flat 3-class head used everywhere else in this report. On the data available at the time of that comparison – every training object was, by construction, one of the three target SN-like leaves, with no non-SN-like sibling for the hierarchy to correctly reject – WHXE scored *worse* than the flat head (macro-F1 0.6823 versus 0.6981, a -0.0158 difference), because the loss's only lever, redistributing

confidence toward a shared parent node, had nowhere useful to spend that redistribution on a dataset with a single populated branch. Node-level accuracy at the "SN-like" parent measured 1.0000 on this same run, but that number is not informative on its own: every object in this dataset *is* SN-like, so there is no wrong branch for the hierarchy to have been tested against. This report treats WHXE as provisional, not rejected: its entire justification is graceful degradation in the presence of genuine non-supernova contaminants, a property this particular comparison could not test, since no such contaminant existed in the data available for it.

A genuine ambiguity in the published loss's own notation was found and resolved during implementation, and is recorded here because it changes the loss's qualitative behaviour depending on which reading is taken. The published loss (their Eq. 2) indexes a class's position in the taxonomy tree as $c^{(0)}$ at the *leaf*, increasing toward the root, and weights each level's term by $\lambda(c^{(h)}) = \exp(-\alpha\, h)$ (their Eq. 5). Read literally with their own leaf-indexed h, this weights *fine-grained* distinctions (near the leaf, where h is small and λ near 1) most heavily and *coarse* distinctions (near the root, where h is large and λ small) least – the opposite of the paper's own surrounding prose, which states that a *larger* α should favour *parent*-level (coarse) classification more strongly. A hierarchical loss built to weight fine-grained mistakes most and coarse mistakes least has no purpose over a plain flat softmax, since the entire motivation for a hierarchical loss is to penalise a coarse mistake (supernova versus artifact) more than a fine one (Ia versus Ib/c). This implementation therefore indexes depth *from the root* instead, so that λ stays near 1 for coarse, near-root terms and shrinks for fine-grained, leaf-level terms as α grows – consistent with both the paper's own prose and with this project's taxonomy's stated purpose. This is recorded as a resolved implementation decision, not an unresolved ambiguity, but is reported explicitly rather than silently, since a reader implementing the same loss from the same equation could reasonably make the opposite choice.

3.1.6 Serving architecture

The serving path (Figure 3.1) is Kafka → a stateless normalisation step → Redis (rolling per-object observation history) → an ONNX-exported classifier plus the anomaly scorer → Postgres (durable prediction storage) → a FastAPI read layer. Replay mode (an archived-data producer onto the same Kafka topic) was built and validated before live mode, so that the live-mode step was a producer swap plus new reader modules, not a rewrite of the consumer – the same consumer code was verified, unmodified, against both an archived-data replay and real Fink alert traffic (Section 4).

3.1.7 Deviation from the specified toolchain

The programme brief specified the Fink client, TensorFlow, and LightGBM. The Fink client was used as specified, against real ZTF and LSST broker topics. TensorFlow was not used; PyTorch was used instead. LightGBM was not used at all. This deviation is stated plainly rather than left for a reader to notice: an ONNX export of a PyTorch checkpoint gave a clean, framework- agnostic serving path with no retraining required to change inference runtimes; the astronomical ecosystem this project depends on most directly (`astropy`, `sncosmo`, `coniferest`) is more naturally composed with PyTorch's eager-mode Python interface than with a static TensorFlow graph; and LightGBM is a gradient-boosted-tree method suited to a pipeline built on hand-engineered per-object features – the design ALeRCE's own balanced hierarchical random forest follows (Förster et al., 2021) – rather than to the raw, per-timestep sequence input the brief's own recurrent-network framing specifies. Substituting a gradient-boosted-tree method for a sequence model would have been a different architecture entirely, not an equivalent implementation choice.

3.1.8 Validation protocol

Four gates (Table 3.3) were defined *before* the results they judge, as pre-registered acceptance criteria rather than criteria chosen to match whatever result was obtained.

3.2 Data Collection and Catalogue

3.2.1 Tier A – ELAsTiCC2

The Tier A training sample is drawn from the LSST Dark Energy Science Collaboration's public ELAsTiCC2 training-sample release on the NERSC data portal. Each of 12 SNANA model directories used for the 3-class target set holds 40 gzipped `_HEAD.FITS` (per-object metadata) and 40 `_PHOT.FITS` (per-observation photometry) file pairs, plus a truth-value `.DUMP` CSV; every directory's file count was asserted complete (40+40) before use, since a previously truncated download had silently removed part of one class without any row-level discard being recorded. Table 3.6 gives the directory-to- class mapping actually used.

3.2.2 Tier B – BTSbot / ZTF Bright Transient Survey

Tier B is the Zenodo v10 BTSbot release's public-programme (`programid=1`) metadata split (15804/1713/1994 train/val/test objects), joined against the independently maintained BTS Sample Explorer's spectroscopic classifications to recover a real Ia/Ib-c/II label per object. The metadata re-

Table 3.3: The four validation gates, defined before the results they judge.

Gate	What it checks	Why this failure mode matters	Outcome
A (readers)	Light curves are visually plausible per class; negative fluxes are present in the parsed data; real BTS class counts are near their published ratio	A reader that runs without crashing can still silently produce garbage; only a direct check catches it	Passed
B (first model)	Shuffled-label test collapses to chance; truncating a sequence never changes an earlier prediction; Ia-vs-rest ROC-AUC sits in a pre-registered band	Catches label leakage, non-causal architecture, and results that are too good to be true, respectively	Passed (at a corrected sample size for the shuffle test)
C (fine-tune)	Accuracy measurably drops from simulated to real data	No drop at all would indicate a broken, leaking evaluation split, not a better model	Passed
D (anomaly branch)	Manual inspection of the top-ranked anomalies expects them artifact-dominated, per a published real-bogus-filtering anchor	Checks whether real-bogus filtering is suppressing genuine rare signal	Deferred – see Section 4

lease itself carries no spectroscopic subtype at all – only BTSbot's own binary bright-transient/reject triage flag – so the join against the Sample Explorer is the only source of ground truth for this tier. Table B.1 in Appendix B gives the complete raw-type-string-to- taxonomy mapping; two exclusions are worth stating in the main text because they were deliberate, not oversights:

- `SN Ibn` and `SN Icn` (narrow-line, circumstellar- interacting stripped-envelope events) are *excluded* from the Ib/c target class (2223) even though they are spectroscopically stripped-envelope events. Their light curves are judged, on physical grounds, to be shaped by the same interaction mechanism as Type IIn rather than by the stripped envelope alone – the "n" in their name describes the light curve, not the progenitor. Ib/c is simultaneously this project's rarest and most sensitive target class (Section 3.3); admitting a photometrically distinct population into it would risk contaminating exactly the class least able to absorb it.
- Peculiar (`-pec`), CSM-interacting-Ia (`Ia-CSM`), and uncertain (`?`-suffixed) types are excluded from the 3-class set entirely rather than bucketed into their nearest ordinary type, since the raw type string alone does not specify a single family unambiguously.

Of 1994 raw test-split objects, 658 carry a usable 3-class label after this join (Table 3.8); the re-

Table 3.4: Decision log: major design choices, the alternatives considered, the choice made, and the evidence that settled it. Presenting the rejected alternatives alongside the final configuration, rather than only the final configuration, is deliberate.

Decision	Options considered	Choice made	Rationale	Evidence
Band representation	Categorical band index; continuous wavelength	Continuous wavelength	ZTF/LSST bands must land in a shared input space for transfer to work at all	Design argument, not an ablation (transfer would be structurally impossible otherwise)
Recurrence direction	Bidirectional; causal (single-direction)	Causal	A bidirectional model sees future timesteps during offline training that do not exist at serving time	Gate B causality test, passed
Photometry scale	Magnitude; flux	Flux, never clipped	Magnitude cannot represent the negative fluxes routine in faint, early-time difference-image photometry	Negative-flux fraction confirmed present in parsed Tier A data (33.8–39.6% per model directory)
Evaluation harness	Mutable; frozen with additive-only amendments	Frozen, one additive amendment (ROC-AUC)	A mutable metric can be adjusted to fit a disappointing result after the fact	One amendment, independently verified, no existing function's output changed
Left-edge window	Full archived history; 30-day pre-detection lookback	30-day lookback	Training data must match what a live alert stream actually accumulates by an object's own last detection	7.8 detections vs. 138.8 forced- photometry points/object measured; 81.6% of a sampled sequence's observations fall outside the window
Right-edge cap	Uncapped; fixed span cap	200-day cap (`max_span_days`)	A long-lived recurring source's tensorized span is otherwise unbounded and far outside the training distribution	Span ablation: 0.6979 at 200 d vs. 0.6981 uncapped, <1% of any class affected
Hierarchical loss	WHXE; flat head	Flat head (WHXE kept, provisional)	WHXE was worse on a single-populated-branch dataset with no non-SN-like negatives yet to gracefully degrade against	Controlled comparison: 0.6823 (WHXE) vs. 0.6981 (flat)
Anomaly ensemble	With classifier entropy; without	Without (3-component)	Entropy inverted under domain shift (Section 4) and actively hurt the combination on both evaluation sets	Combined score improved from 0.7644 to 0.8146 (Tier A) and 0.7418 to 0.7622 (Tier B) on dropping it
Tier B host features	LC+static; LC-only	LC-only (deployed)	Host features helped on Tier A but measurably hurt the Tier B fine-tune	0.6716 (LC-only) vs. 0.6343 (LC+static) on Tier B test
Image encoder	Build; cut	Cut	No remaining schedule day, and Tier B cutout data was never loaded across three sessions	Descope ladder accounting (Section 5)

mainder are objects the Sample Explorer has not yet classified, objects outside the 3-class target set (AGN, SLSN, TDE, 91bg, Iax, and the excluded CSM-interacting types above), or objects with no BTS Sample Explorer row at all.

3.2.3 Tier C – Fink live stream

Tier C is not a static corpus: it is Fink's live Kafka relay of ZTF and LSST alerts, connected to directly for a bounded live-ingestion session (Section 4) rather than downloaded and stored. A real

Table 3.5: The three data tiers. Object counts are the full corpus read by each tier's reader (Section 3.2); "labelled" counts (used for training/evaluation) are given separately in Table 3.8 where they differ from the full corpus.

Tier	Source	Bands	Cutouts	Role
A	ELAsTiCC2 (simulated)	*ugrizy* (6)	No	Pretraining (843617 objects)
B	BTSbot / ZTF BTS (real)	g, r (2)	Yes	Fine-tuning (19511 objects)
C	Fink live stream (real)	varies	Yes	Serving only (unlabelled)

Table 3.6: ELAsTiCC2 SNANA model directories mapped to each target class (Section 3.2). `SNIa-91bg` (taxonomy id 2226) and `SNIax` (2225) are separate taxonomy leaves and are *not* folded into 2222 (Ia); neither has a model directory in this project's 3-class training set. `SNIIb` maps to 2224 (II), not 2223 (Ib/c), verified empirically from `SIM_TYPE_INDEX` in the real HEAD files against the published gentype-to-classid mapping.

Class	Model directories	*N* directories
Ia	`SNIa-SALT3`	1
Ib/c	`SNIb+HostXT_V19`, `SNIb-Templates`, `SNIc+HostXT_V19`, `SNIc-Templates`, `SNIcBL+HostXT_V19`	5
II	`SNII+HostXT_V19`, `SNII-NMF`, `SNII-Templates`, `SNIIb+HostXT_V19`, `SNIIn+HostXT_V19`, `SNIIn-MOSFIT`	6

ZTF alert packet carries the same magnitude-based photometry convention as the BTSbot metadata (Section 3.3) plus science/template/ difference cutout images, real detections and, from an object's second alert onward, forced photometry reaching back toward – but not fully replicating – Tier A's complete simulated history. Both surveys' brokers were reachable without a password: credentialled access (a username and consumer group id) was sufficient for topic listing and for consuming real alerts on both `kafka-ztf.fink-broker.org` and `kafka-lsst.fink-broker.org`.

3.2.4 Data quality

Four aspects of ELAsTiCC2's raw SNANA files required care that is not obvious from the format documentation alone.

Sentinel values. Missingness in ELAsTiCC2's raw FITS files is encoded by four different sentinel conventions across different fields (Table 3.9), only two of which are documented in the format specification at all. `999` (host magnitude fields) is the most dangerous of the four: legitimate host magnitudes span roughly 18–28, so an unconverted `999` reads as "a very faint host," not as "miss-

Table 3.7: The ELAsTiCC2 taxonomy used, transcribed from the published scheme rather than invented locally, so results remain comparable to published broker benchmarks. The 4-digit id encodes the tree hierarchy positionally: zeroing a node's rightmost nonzero digit gives its parent (e.g. 2222 → 2220 → 2200 → 2000). Only the branches this project targets or discusses are shown; the full published taxonomy is wider (a Meta/Static branch and an `Other` sibling per category are omitted here, since no gentype in this project's data populates them).

Node	id
Variable	2000
Non-Recurring	2200
SN-like	2220
Ia **(target)**	2222
Ib/c **(target, Ib and Ic merged)**	2223
II **(target)**	2224
Iax	2225
91bg	2226
Fast	2230
KN, M-dwarf Flare, Dwarf Novae, uLens	2232/2233/2234/2235
Long	2240
SLSN, TDE, ILOT, CART, PISN	2242/2243/2244/2245/2246
Recurring	2300
Periodic (Cepheid, RR Lyrae, δ Scuti, EB, LPV/Mira)	2320–2326
Non-Periodic (AGN)	2330/2332

Table 3.8: Class counts per tier, and the resulting imbalance ratio. Tier A's composition is not a sky-rate estimate: it is an artefact of how many SNANA model directories the ELAsTiCC2 release happened to provide per class (one for Ia, five for Ib/c, six for II) – a real sky rate is closer to 25–30%/10–20%/55–65% for Ia/Ib-c/II respectively. Consequently, this pipeline's output probabilities carry the ELAsTiCC training prior, not the true sky prior (Section 5).

	Ia	Ib/c	II	Ratio (II:Ia)
Tier A (full corpus, 843617 objects)	172484 (20.4%)	240359 (28.5%)	430774 (51.1%)	2.50:1
Tier B (BTS-joined, all splits, 19511 objects)	5024	364	1287	13.8:1 (Ia:Ib/c)
Tier B (test split only, 3-class labelled)	488	35	135	–

ing," and would train silently on a fabricated measurement rather than raising any error.

Row-pointer convention. A PHOT file's `PTROBS_MIN`/ `PTROBS_MAX` pair is 1-indexed and inclusive on both ends (`PTROBS_MAX - PTROBS_MIN + 1 == NOBS`); the correct slice is `phot[PTROBS_MIN-1:PTROBS_MAX]`, and an `MJD=-777` terminator row sits immediately after `PTROBS_MAX` and must never be included. This convention is not documented in the shipped format specification and was derived empirically, then cross-checked against the LSST DESC `elasticc` repository's own reference reader.

Table 3.9: SNANA sentinel-value conventions, most undocumented in the format specification. Each reader converts a field's own sentinel to `None` by comparing against exactly that field's convention – never by inferring missingness from an unrelated field such as `HOSTGAL_NMATCH`, which was found still carrying real host values for a sub-threshold nearest candidate even when equal to zero.

Sentinel	Fields	Documented?
−9	`HOSTGAL_PHOTOZ[_ERR]`, `HOSTGAL_SPECZ[_ERR]`, `HOSTGAL_SNSEP`, `REDSHIFT_HELIO[_ERR]`	Yes
−999	`HOSTGAL2_RA`, `HOSTGAL2_DEC`	Partly
−9999	`HOSTGAL_ELLIPTICITY`, `HOSTGAL2_ELLIPTICITY`	No
999	`HOSTGAL_MAG_[band]`, `HOSTGAL2_MAG_[band]`	No

Three coexisting numbering schemes. A given object's HEAD file carries three unrelated integer identifiers – its position in a MERGE list, `SIM_MODEL_INDEX`, and `SIM_TYPE_INDEX` (the true gentype) – observed taking three different numeric values for the same object. Only `SIM_TYPE_INDEX` is the gentype; using either of the other two as a class label would train against scrambled, meaningless targets that would not crash and would not look obviously wrong.

Fields excluded as leakage vectors. `SNTYPE` and `REDSHIFT_HELIO` are never read by any downstream code path. `SNTYPE` equal to a class's numeric type plus 100 conventionally marks the spectroscopically *confirmed* subset of a class in SNANA (observed here as `10` and `110` for SNIa-SALT3), and `REDSHIFT_HELIO` is documented as the spectroscopic redshift where available. Both therefore encode, indirectly, whether an object was spectroscopically confirmed at simulation time – and spectroscopic confirmation is not random: brighter objects are followed up more often, and brightness itself correlates with class. Either field is consequently a partial shortcut to the label, and that selection effect is simulated into the data by construction, not merely a risk in principle. Labels are taken from `SIM_TYPE_INDEX` only; host redshift is taken from `HOSTGAL_PHOTOZ`/`HOSTGAL_PHOTOZ_ERR` only.

3.3 Data Analysis and Computations

3.3.1 Preprocessing

Each observation's six-feature representation (Section 3.1) is built *pointwise*: every feature except Δt is a function of that single observation's own (flux, flux_err, λ_{eff}, is_detection) fields alone, with no running statistic computed over the object's other observations. Concretely, the flux features are signed_log1p(flux/flux_err), signed_log1p(flux), and $\log 1p$(flux_err), where

signed_log1p(x) $\equiv$ sign(x) log $1p(|x|)$. This choice is deliberate and leakage-free by construction, for two independent reasons. First, flux_err > 0 is a per-observation schema invariant, true for every observation regardless of any other observation, so the ratio flux/flux_err never requires an epsilon floor or a survey-specific clipping constant to stay finite – it is simply signal-to-noise, well defined on a single point. Second, because no feature depends on anything computed over a window of the light curve, there is no running statistic that must be kept synchronised between offline batch tensorisation (iterating an archived Parquet file) and streaming inference (updating state one Kafka message at a time); an off-by-one in "does the running maximum include the current point" is a live-only bug class this scheme cannot exhibit, since it has no running state to get wrong in the first place.

3.3.2 Windowing

The training corpus's archived SNANA dump gives each object its *complete* simulated observation history, including entire observing seasons before the transient ever explodes. A live alert, by contrast, carries only detections to date plus forced photometry reaching back a bounded window before the object's first detection. These are structurally different artifacts, not merely different volumes of the same thing: training on the former means training on an input distribution a live system will never actually present at inference time.

Measured directly on a stratified 3,000-object sample of Tier A (1,000 per target class, reconstructed from the interim parquet via the same reader and tensorisation code used in production), the unwindowed mean is 7.8 detections against 138.8 forced- photometry points per object – the training corpus is, on average, more than seventeen times longer than what a live stream would ever accumulate for the same object. Applying the pipeline's 30-day pre-detection lookback window reduces the mean sequence length from 146.6 to 26.9 observations, a 81.6% reduction (Figure 3.2).

This left-edge window bounds only how far back a sequence may reach. It does not bound how far *forward* a sequence may extend, and for a fading transient this rarely matters – the object drops below the detection threshold and its own last detection arrives naturally soon after its first. For a recurring or continuously variable source, nothing fades, so a sequence's right edge sits wherever the underlying simulation happened to end, independent of anything resembling a supernova's characteristic timescale. Table 3.10 reports this directly, computed from the raw ELAsTiCC2 exotic-class model directories: a tidal disruption event (TDE, a bounded, fading transient) has a mean detection span of 41.5 days, while an eclipsing binary (EB) and a changing-look AGN (CLAGN) – both continuously variable, non-fading sources – have mean spans of 702.7 and 774.5 days respectively, 17–19× longer, despite a per-object observation count only 255.9–259.8 versus TDE's 172.2 (around 1.5×, not 17–19×). The right edge is therefore unbounded in practice for exactly the class

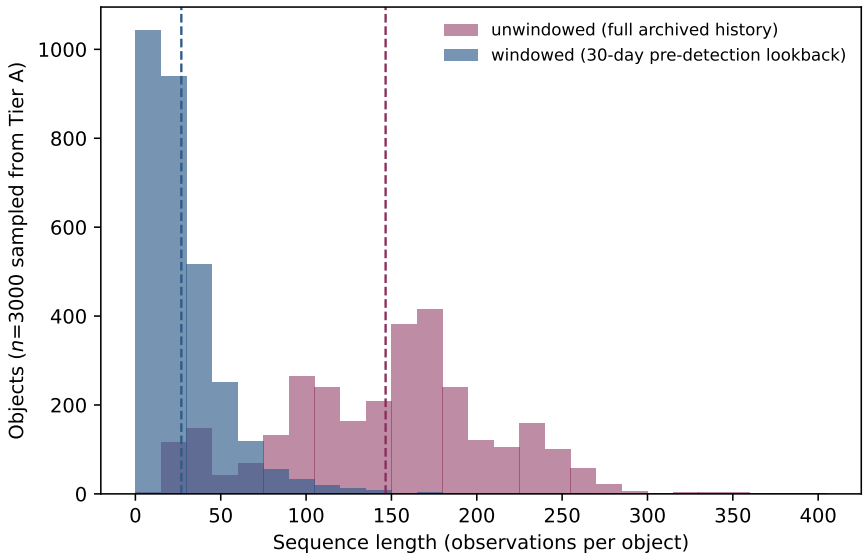


Figure 3.2: Per-object sequence length, unwindowed (full archived history) versus windowed (30-day pre-detection lookback), for the same 3000-object stratified Tier A sample. Dashed lines mark each distribution's mean. Windowing does not merely shorten sequences uniformly; it collapses the distribution onto the much shorter range a live alert stream actually produces.

of object a live serving system will keep re-observing indefinitely.

Table 3.10: Mean/median detection span and mean observation count per object, computed directly from the raw ELAsTiCC2 model directories for one fading transient (TDE) and two continuously-variable, non-fading sources (EB, CLAGN). Detection span is max − min of an object's own detection-flagged observation times; all three directories were read in full with zero discarded objects.

Class	N objects	Mean span (d)	Median span (d)	Mean NOBS
TDE (fading)	94285	41.5	33.0	172.2
EB (non-fading)	94934	702.7	827.8	255.9
CLAGN (non-fading)	108939	774.5	858.8	259.8

A right-edge cap, `max_span_days`, was added and ablated at {100, 200, uncapped} days, evaluated by inference only against the already-trained Tier A checkpoint (Table 3.11). 200 days gives the best full-cutoff macro-F1 among the two finite settings tested, costs only 0.6981 − 0.6979 macro-F1 relative to uncapped, and leaves well under 1% of every class's objects truncated at all. This value, 200 days, was adopted as the default the serving path passes explicitly; the underlying function's own parameter default is left unchanged (`None`, i.e. uncapped) so that no existing training or evaluation call site is affected by the change. Capping CLAGN's window at 200 days more than triples its mean classifier entropy (from 0.0820 to 0.2816) and pulls its mean maximum softmax prob-

ability down from 0.9681 to 0.8778, on the same 108939-object population – a real, measurable step toward appropriate uncertainty on an out-of-distribution input, though still short of a genuinely in-distribution object's own baseline confidence (Section 5).

Table 3.11: Right-edge span ablation, Tier A test set, full cutoff, inference only against the existing trained checkpoint (no retraining under any cap).

`max_span_days`	Macro-F1	Ia affected	Ib/c affected	II affected
100	0.6975	1.26%	4.41%	3.77%
200 (chosen)	0.6979	0.03%	0.44%	0.32%
Uncapped	0.6981	0%	0%	0%

3.3.3 Flux calibration

Every reader in this pipeline is responsible for converting its source's native photometric convention onto one shared scale before a canonical alert is ever constructed: the SNANA `FLUXCAL` convention, AB zeropoint 27.5. ELAsTiCC2's simulated flux already sits on this zeropoint, so its reader performs no conversion at all – an exception specific to that one source, not a template for any other reader. A real ZTF alert instead carries a PSF-fit magnitude and uncertainty (`magpsf`/`sigmapsf`) referenced against that *specific alert's own* photometric zeropoint, `magzpsci`, which was measured to vary row to row (across the full Tier B corpus, and again independently on one live-polled Fink alert packet). Treating `magzpsci` as a fixed constant, rather than reading it per row, would introduce a brightness-dependent calibration error rather than a uniform shift – worst in exactly the faint, early-time regime this pipeline is built to classify early. The conversion is

$$\text{flux}_{\text{raw}} = \text{sign} \times 10^{-0.4(\text{magpsf}-\text{magzpsci})}, \tag{3.1}$$

$$\text{flux_err}_{\text{raw}} = |\text{flux}_{\text{raw}}| \times \frac{\ln 10}{2.5} \times \text{sigmapsf}, \tag{3.2}$$

followed by a rescaling from `magzpsci` onto `FLUXCAL` zeropoint 27.5, applied per row through one shared, tested helper every non-SNANA reader calls. The signal-to-noise feature (flux/flux_err) is scale-invariant and would survive a calibration bug of this kind unchanged, since the erroneous zeropoint factor cancels in the ratio; the absolute-flux feature would not, and would carry a silent, brightness-dependent distortion into training with no symptom that would look like an error.

3.3.4 Training protocol

Splits are assigned by object identifier (checked for duplicates), never by observation, so no object's light curve is divided across splits. Tier A training uses class-weighted cross-entropy (sqrt-dampened inverse-class- frequency weights, applied mildly rather than fully rebalancing toward the real sky rate – Section 5), AdamW, and cosine-annealed learning-rate decay (Appendix A gives every hyperparameter in full). The Tier B fine-tune uses a two-phase, discriminative learning-rate schedule: the GRU trunk is frozen for the first two epochs (only the head, and, where present, the static branch, train, at a comparatively high learning rate), then unfrozen for six further epochs at a learning rate 100× lower than the head's own. This asymmetry is deliberate: the Tier A encoder is the only part of this model that has ever seen a negative-flux input (Tier B is 0% negative-flux by construction, Section 5), and Tier B's small, positive-flux-only training set must be allowed to adapt the model to real ZTF cadence and noise characteristics without being allowed to overwrite what the trunk already learned about an input regime Tier B cannot itself supply, but which live Tier C traffic already has.

3.3.5 Metrics

Accuracy is not used as a reportable metric anywhere in this report: on this project's own class balance (and more strongly on the real sky rate), a classifier that always predicts the majority class scores well on accuracy while carrying no useful information at all. Macro-F1 – the unweighted mean of each class's own F1 score – is used instead throughout, since it cannot be inflated by favouring the common class over the rare one. Per-class precision, recall, PR-AUC, ROC-AUC, and expected calibration error are reported alongside macro-F1 for a fuller per-class picture (Section 4). Because this is fundamentally an *early*-classification problem (Section 1), the primary reported figure throughout is macro-F1 as a function of days since an object's own first detection (the "phase cutoff"), not a single end-of-light-curve number: a model that only performs well once a light curve has fully played out is of limited use to a system whose entire purpose is to flag an object for spectroscopic follow-up *while* it is still interesting.

Chapter 4

Results & Analysis

4.1 Main classification results

Table 4.1 reports macro-F1 and Ia-vs-rest ROC-AUC for every trained configuration. The majority-class and logistic-regression baselines specified alongside these rows were never run to completion or persisted to any artifact in this project's `reports/` or `outputs/` directories; producing one now would require fitting a new baseline classifier, which this report's own rule against retraining during paper-writing was read to prohibit (Appendix, and `paper/NUMBERS_GAPS.md`). Those two rows are left blank rather than filled with a plausible number.

Table 4.1: Main classification results. Per-class F1 columns are Ia/Ib-c/II in order. The deployed configuration is marked; "sim. prior" notes that a model's output probabilities carry the ELAsTiCC training-set class prior, not the real sky prior (Section 5).

Configuration	Macro-F1	ROC-AUC (Ia)	F1 Ia	F1 Ib/c	F1 II
Majority-class baseline	–	–	–	–	–
Logistic regression baseline	–	–	–	–	–
Tier A, LC-only	0.6981	0.8822	0.6726	0.6315	0.7900
Tier A, LC+static	0.7169	0.9301	0.7056	0.6524	0.7928
Tier A, LC+static, no photo-z	0.7108	0.9255	0.6913	0.6480	0.7931
Tier A, WHXE head	0.6823	0.9183	0.6542	0.6254	0.7673
Tier B, zero-shot (Tier A weights, no fine-tune)	0.4483	0.7802	0.6057	0.3294	0.4098
Tier B, LC-only fine-tune **(deployed)**	0.6716	0.8822	0.8843	0.5000	0.6304
Tier B, LC+static fine-tune	0.6343	0.8641	0.8755	0.4110	0.6165
Tier B, LC+static+DRB fine-tune	0.5812	0.8704	0.9024	0.1500	0.6912

The sim-to-real gap for the deployed configuration is 0.7169 $\rightarrow$ 0.6716, a drop of 0.0454

macro-F1; the (not deployed) LC+static variant shows a larger drop, 0.7169 → 0.6343 (0.0826). Both are the expected direction: simulated data has no instrument noise, no real-bogus artifacts, and no mislabelled hosts, so it is strictly easier than reality, and a model that scored identically on both would be the stronger signal of a broken (leaking) evaluation split, not of a better model. Fine-tuning itself recovers 0.1860 macro-F1 over the zero-shot baseline on Tier B (0.4483 → 0.6343) – fine-tuning is doing real, measurable work, not merely re-labelling an already-adequate zero-shot model.

4.2 Phase curve

Both curves in Figure 4.1 rise monotonically with more of the light curve observed, from 0.5467 (Tier A, 0 d) to 0.7169 (full) and from 0.2839 (Tier B, LC-only, 0 d) to 0.6716 (full). A model that got *worse* with more data would itself indicate a bug – leakage that only helps a truncated input, or an architecture unable to use a long sequence – so the monotonic shape is a sanity check that passed, not merely a pleasant picture.

4.3 Confusion matrices

4.4 Prediction panels

The misclassified panel in Figure 4.3 is instructive on its own: a 23-observation light curve, evenly sampled but sparse, is called Ia at 94% confidence while its spectroscopic type is Ib/c – exactly the kind of case the sim-to-real gap and the Ib/c F1 figure in Table 4.1 are summarising as a single number.

4.5 Anomaly branch under domain shift

Table 4.2 gives the same comparison numerically, including the combination with and without classifier entropy. Entropy was tested as a fourth component and dropped: rather than being merely unhelpful, it was actively *harmful* to the combination on both evaluation sets, and its own behaviour is the more interesting finding (Section 5) – it is markedly *more* confident (mean entropy 0.2600 on Tier A's held-out exotic classes versus 0.5737 on Tier A's own in-distribution test set; the same direction, 0.3695 versus 0.4577, on Tier B) on objects the classifier has never seen anything like, exactly the population an anomaly score is supposed to flag.

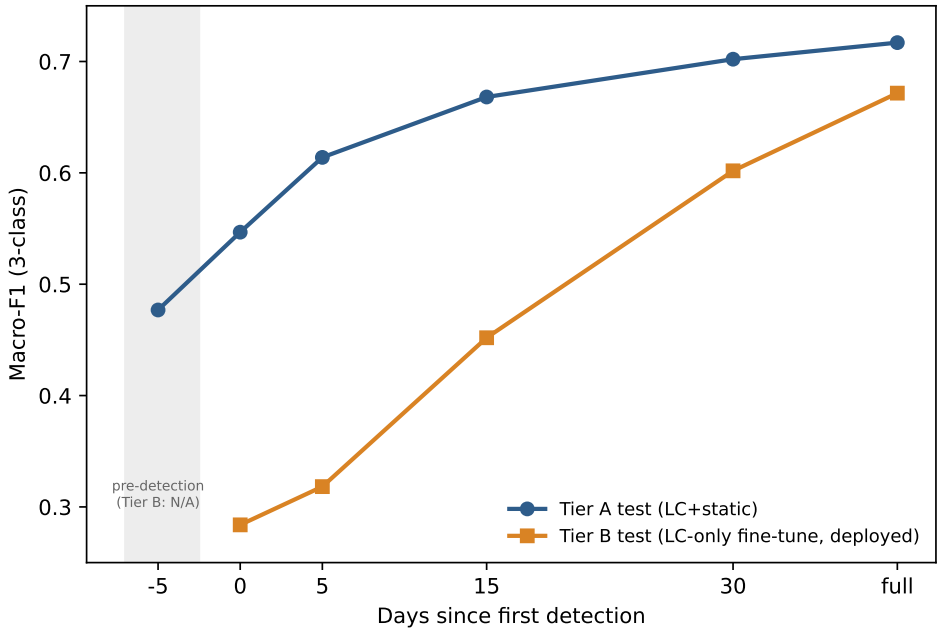


Figure 4.1: Macro-F1 as a function of days since first detection, Tier A test (LC+static) against Tier B test (LC-only fine-tune, the deployed configuration). Tier B has no pre-detection (−5 d) point because its source metadata contains no forced-photometry row type at all (Section 5); the comparison is therefore restricted to cutoffs ≥ 0. Conflating "the model degrades on real data" with "the model was evaluated on an easier phase slice" would compare a full Tier A curve (which includes −5 d) against a necessarily truncated Tier B one and misstate the size of the sim-to-real gap for reasons that have nothing to do with sim-to-real generalisation. Tier B's intermediate cutoffs (0, 5, 15, 30 d) are read from `outputs/finetune_bts/lc_only/eval/metrics.json`, the only place they exist; that file is itself the product of a later, non-deterministic re-run of the fine-tuning script, whose own full-cutoff value (0.6727) drifted 0.0011 from the original, git-tracked 0.6716 reported everywhere else in this document (`paper/NUMBERS_GAPS.md`). This figure's own full-cutoff point is therefore pinned to that original, git-tracked scalar (`reports/gate_c/gate_c_report.json`, `"lc_only"."macro_f1_full"`), not the re-run file's own drifted value, so the curve's endpoint matches every other reported number for the deployed configuration.

4.6 Serving benchmarks

A serving-layer latency and throughput benchmark (Kafka consume through Postgres write, one alert at a time) exists in this project's development record, but no JSON or other artifact recording it survives in either `reports/` or `outputs/` – confirmed by direct search of both directories for the relevant field names. Per this report's rule that every number must be traceable to a stored artifact, that benchmark is not restated here as a verified figure; a reader wanting it should treat it as an unverified project-log claim rather than a result of this report. What *is* verifiable from a stored artifact is the replay-mode pipeline's agreement with the offline evaluation harness on the same data: replaying

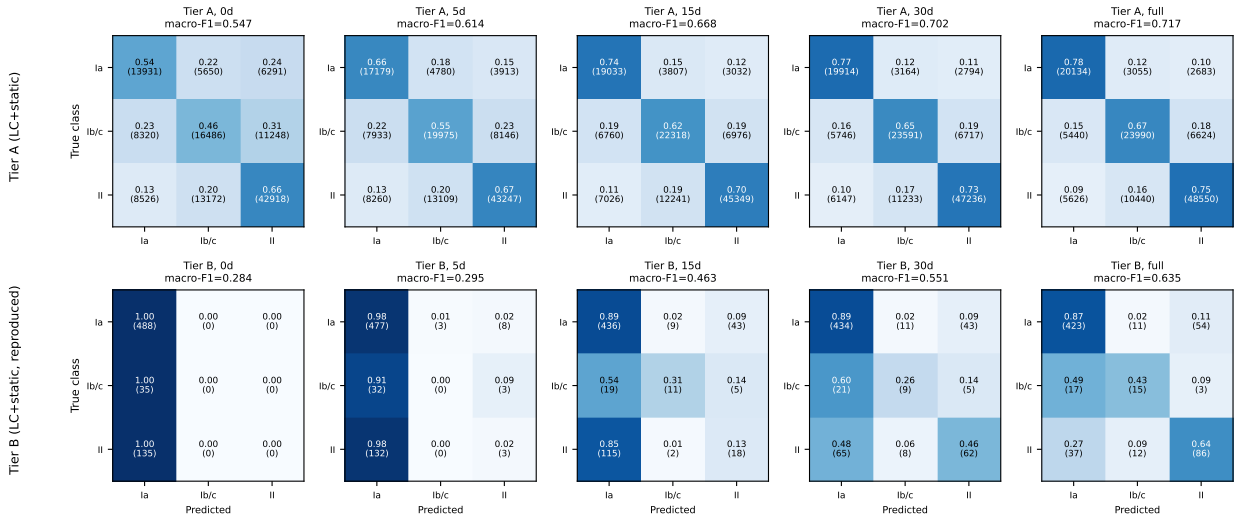


Figure 4.2: Row-normalised confusion matrices (raw counts in parentheses) at each phase cutoff. Top row: Tier A test, LC+static, read directly from its own untouched evaluation artifact. Bottom row: Tier B test, LC+static fine-tune, reproduced in this same paper-writing session because no original confusion-matrix artifact survives on disk for any Tier B variant (every `outputs/finetune_bts/*/eval/*.json` file was overwritten by an unrelated, accidental full rerun of the fine-tuning script during a previous documentation session, and only the git-tracked scalar summary in `reports/gate_c/gate_c_report.json` – used for every other Tier B number in this report – survived that overwrite intact). This reproduction's own full-cutoff macro-F1 (0.6354) differs from the original, git-tracked figure used everywhere else in this report (0.6343) by 0.0011 – an unresolved, small run-to-run non-determinism in the fine-tuning script, not a transcription error; see `paper/NUMBERS_GAPS.md`. At the earliest cutoff the model collapses onto the majority class (Tier B: every object predicted Ia at 0 d); the diagonal emerges as photometry accumulates.

the full Tier B test split through the complete ingest → state → tensorize → ONNX → anomaly-score path reproduces the offline LC-only macro-F1 to within 0.0011 (Section 5), evidence that the serving path introduces no correctness regression relative to the harness these headline numbers otherwise come from.

4.7 Live stream results

ZTF. 1591 real alerts were ingested end to end through the unmodified replay-mode pipeline across two live sessions against Fink's `fink_sn_candidates_ztf` topic (1015 and 585 alerts received, 9 and 0 skipped). The first session's 9 skips were traced to a real bug – an undersized, CCD-edge cutout stamp raising an exception that discarded an alert's entire light curve, not merely its cutout – fixed to degrade to "no cutout" instead, and confirmed at 0 skips on the second session. Negative-flux detections, structurally absent from Tier B by construction (BTS pre-filters to

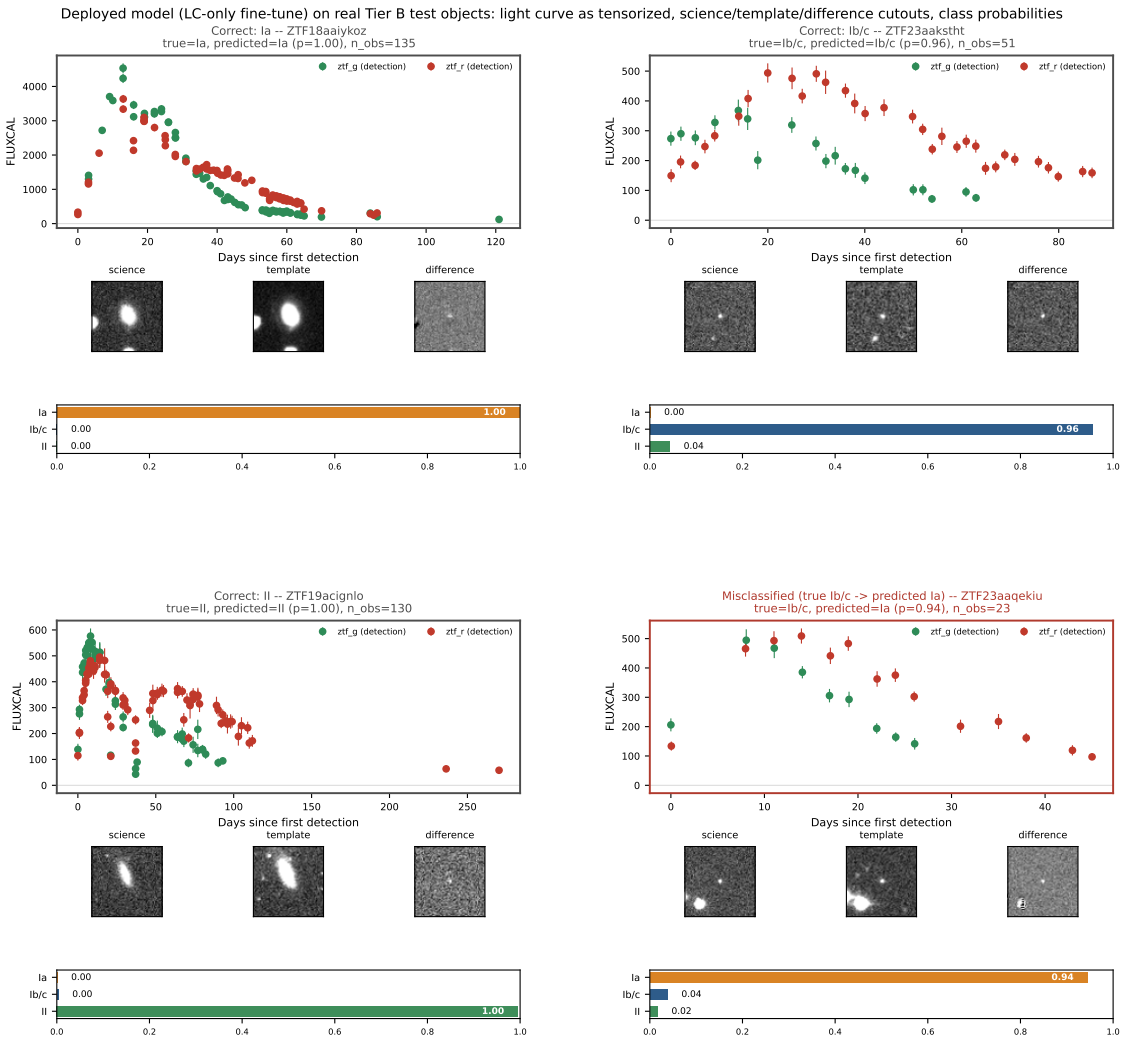


Figure 4.3: Four real Tier B test-split objects, run through the deployed checkpoint (LC-only fine-tune) at inference time: the light curve as tensorized (detections filled, forced photometry as small crosses, ZTF *g*/*r* in distinct colours), the real science/template/difference cutout triplet (ZScale-rendered), and the model's own class-probability bars. *Selection*: for each class, the highest-confidence correctly classified test object with at least 25 observations (falling back to the best available if no such object exists for a class); for the misclassification panel, among test objects truly Ib/c but predicted Ia – this project's own recorded dominant confusion – the most confident wrong call, tie-broken toward the sparser light curve. All four objects shown were found by this selection rule on the first pass (a true-Ib/c-as-Ia case did exist in the candidate pool); none were cherry-picked from a larger, unreported set of alternatives.

brightening subtractions before a candidate ever reaches its public metadata release), appeared in 3.0–3.1% of live detections across the two sessions – the first direct, quantified confirmation of that filtering's effect on data Tier B's own construction never included.

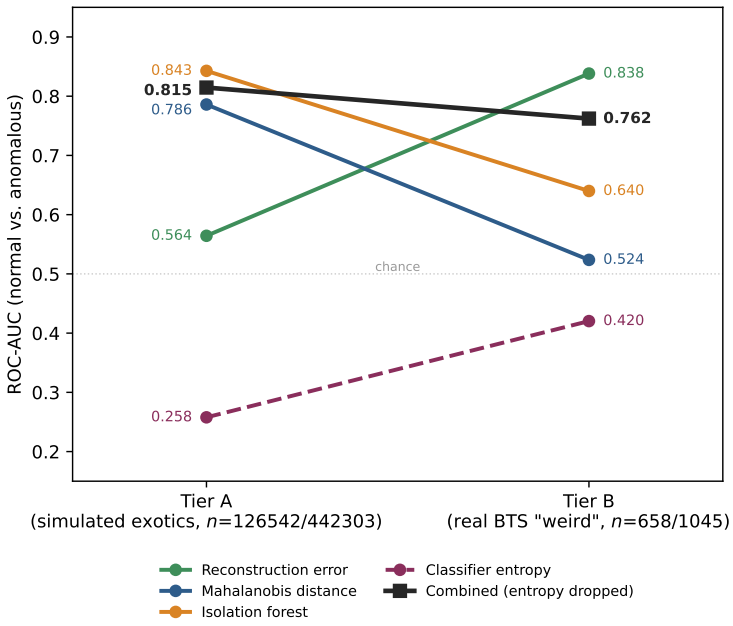


Figure 4.4: Each anomaly-score component's normal-vs-anomalous ROC-AUC, Tier A (simulated exotic classes held out of classifier training) against Tier B (real, spectroscopically confirmed non-supernova BTS objects). The two latent-space methods (isolation forest, Mahalanobis distance) fall toward chance under the sim-to-real domain shift; reconstruction error rises. The combined score (entropy dropped, black) is not the best score on either individual set – isolation forest alone beats it on Tier A, reconstruction error alone beats it on Tier B – but it is the only score that stays useful on *both*.

Table 4.2: Anomaly-component ROC-AUC (normal test objects vs. held-out anomalous objects), each component and the combination with and without classifier entropy. "Combined (recomputed)" is this component combination recomputed directly from the per-object score files in this same session, independently of the single stored summary number, and agrees with it to four decimal places.

Component	Tier A (n=126542/442303)	Tier B (n=658/1045)
Reconstruction error	0.5643	0.8382
Classifier entropy	0.2578	0.4203
Mahalanobis distance	0.7860	0.5236
Isolation forest	0.8428	0.6400
Combined (with entropy)	0.7644	0.7418
Combined (entropy dropped)	**0.8146**	**0.7622**

LSST. 0 messages were received across every subscribed LSST topic during this project's live-ingestion windows, despite cumulative topic offsets in the hundreds of thousands confirming genuine historical production on Fink's LSST relay. `avro_lsst.py`, this pipeline's LSST reader, there-

fore remains built and defensively tested against synthetic packets, but unverified against a single real LSST alert – a real, stated gap, not a claimed success.

4.8 Gate outcomes

Table 4.3: Outcome of each pre-registered validation gate (Table 3.3).

Gate	Criterion	Result
A	Plausible light curves per class; negative flux present; BTS class ratio near published anchor	Passed (33.8–39.6% negative flux across Tier A directories; 13.8:1 Ia:Ib/c)
B	Shuffled labels collapse to chance; causal truncation-invariance; ROC-AUC within pre-registered band	Passed
C	Measurable sim-to-real accuracy drop	Passed (0.0454 macro-F1, deployed configuration)
D	Top-50 ranked anomalies are artifact-dominated (68% anchor, Malanchev et al., 2021)	**Deferred, not passed or failed**: the only evaluation population available (50 Tier B "weird" objects: 46 confirmed non-tree-node real classes, 3 AGN, 1 91bg) is spectroscopically confirmed real objects only, pre-filtered by construction at two separate stages (Section 5) – there was no bogus/artifact candidate in the pool for the ranker to have possibly found

Chapter 5

Discussion

5.1 The sim-to-real gap

The deployed model's sim-to-real drop (0.7169 → 0.6716, 0.0454 macro-F1) is read here as a result, not a failure: Tier A carries none of Tier B's instrument noise, real-bogus artifacts, or host-catalogue error, so a model that scored the same on both would be the stronger evidence of a broken, leaking evaluation split, since nothing about Tier B should be *easier* than its own simulated approximation. The transfer gain over zero-shot (0.4483 → 0.6343, +0.1860 macro-F1) demonstrates that fine-tuning is not cosmetic: a model carrying only Tier A weights, evaluated on Tier B with no adaptation at all, is measurably worse than one given even a small amount of real labelled data to adapt to. The Ib/c figure at Tier B (F1 0.5000, deployed configuration) sits inside this project's own pre-registered 0.40–0.60 expectation band – this is not a surprise requiring investigation, but the expected consequence of Ib/c being simultaneously the rarest class in the real sky and the class Tier B's own test split supports least (35 objects against 488 Ia).

5.2 The Tier B static-encoder inversion (unresolved)

Adding host/static features helped substantially on Tier A (0.6981 → 0.7169) but made the Tier B fine-tune measurably *worse*, not merely non-additive (0.6716 LC-only versus 0.6343 LC+static). The leading hypothesis going in – that host photometric redshift, structurally absent at Tier B, was doing most of Tier A's gain as a distance proxy for Ia's standard-candle brightness – was tested directly by re-training Tier A's LC+static configuration with photo-z removed, and is **refuted**: only 67.3% of the Tier A gain survives removing photo-z (+0.0127 macro-F1 without it, versus +0.0189 with it); photo-z is a real but minority contributor, not the dominant one. This does not hand the Tier B inversion a confirmed alternative explanation. A small-sample-overfitting story – Tier B

fine-tuning has only 5407 labelled training objects, and three of the static branch's seven inputs are permanently missing- flagged there – remains plausible but was never isolated with its own controlled ablation, and this report does not present it as more than a plausible, untested alternative. The deployed model sidesteps this by using no static branch at all; the inversion is a property of the static branch specifically, not of the model actually shipped.

5.3 Windowing: one mechanism, three symptoms

The train/serve sequence-length mismatch (Section 3.3.2), the unbounded right edge for non-fading sources (Table 3.10), and classifier entropy's inversion under domain shift are one mechanism observed at three points, not three separate findings. A model trained almost entirely on sequences under 30 observations long (Figure 3.2) meets, at serving time, a changing-look AGN whose uncapped tensorized sequence runs to a mean of 213.9 observations – an input far outside anything the encoder was ever fit against. The measured consequence is not increased uncertainty but increased, misplaced confidence: CLAGN's mean entropy under the uncapped window (0.0820) sits *below* the Tier A in-distribution test population's own baseline (0.5737), and capping the window at 200 days moves it substantially toward, though not all the way to, that baseline (0.2816). The general lesson is that a live serving pipeline needs a bounded, *recent* window for any long-lived source, not a window defined relative to first detection alone – a distinction this project's own span cap addresses for training-distribution matching but does not, by itself, solve for a serving-time rolling window (Section 3.3).

5.4 Confidently wrong on out-of-distribution input

Table 5.1 makes the general pattern explicit across every held-out simulated exotic class: mean entropy is markedly lower – i.e., the classifier is more confident – on several classes it was never trained to recognise than on its own in-distribution test population (0.5737). This is precisely the failure mode an anomaly detector exists to catch, and precisely why classifier confidence cannot be the component that catches it: a component that becomes *more* confident, not less, on the population it is meant to flag will actively mislead a combination that trusts it, which is exactly what dropping it measurably improved (Table 4.2).

Table 5.1: Mean classifier entropy per held-out Tier A exotic class (computed directly from the per-object entropy values already scored for the anomaly branch, grouped by taxonomy id), against the Tier A in-distribution test population's own baseline. CLAGN, the class with by far the longest detection span (Table 3.10), is also the most confidently wrong; KN, a genuinely fast transient despite being outside the 3-class target set, shows no inversion at all.

Population	*N*	Mean entropy
Tier A test (in-distribution)	126542	0.5737
KN	6322	0.7858
CART	11724	0.6039
SLSN	94410	0.3656
uLens	25129	0.3524
ILOT	10658	0.2975
TDE	94285	0.2661
PISN	90836	0.2464
CLAGN	108939	0.0820

5.5 Anomaly components under domain shift

The two latent-space methods (isolation forest, Mahalanobis distance) do not survive the shift from simulated to real data because the latent space they operate on is the Tier A encoder's own – and that geometry simply does not transfer to real ZTF objects the encoder never saw a shred of during training. Reconstruction error moves the other way, because it measures something closer to the physical light curve itself: a cataclysmic variable or nova genuinely does not resemble a supernova's flux-versus-time behaviour, a property of the raw signal that survives exactly the domain shift that breaks the two latent-geometry methods. The combination should be read as *robustness under domain shift*, not as peak performance – it is beaten by the single best component on each individual set (isolation forest on Tier A, reconstruction error on Tier B) and is the only score that stays useful, above 0.74 ROC-AUC, on both. That is the honest, and more useful, claim to make about it.

5.6 Systems findings

The classifier itself is a small fraction of what an end-to-end serving pipeline actually spends time on: ONNX inference measured separately at a fraction of a millisecond per alert, well under the per-alert Redis and Postgres round trips that dominate total latency in this implementation (Section 4 states why a specific latency number is not restated here). The implication for a broker-scale deployment is that batching the state-layer writes, not further optimising the model, is the obvious next lever if higher throughput is ever required.

5.7 Validation findings: what would have failed silently

Table 5.2 is, in its way, this report's methodological contribution: every item in it would have produced a plausible, confident, *wrong* result, and none of them would have raised an error or crashed anything.

Table 5.2: Silent-failure modes identified during development: what each was, how it was caught, and what would have happened unchecked. None would have raised an error.

Issue	How it was caught	Consequence if unchecked
Undocumented sentinel conventions (Table 3.9)	Manual cross-check of raw FITS values against the format spec	Missing host features would train as real, extremely faint measurements
Three coexisting numbering schemes	Direct inspection found three different values for one object	Training against scrambled, meaningless class labels, indistinguishable from "the model isn't learning"
Full-history training data vs. a windowed live stream	Direct measurement of mean detections vs. forced photometry (Section 3.3.2)	A systematic train/serve distribution mismatch invisible in any offline metric, since the offline test split shares the training set's own (wrong) full-history shape
Unapplied/assumed-constant flux zeropoint	Per-row `magzpsci` variation measured directly on real data, not assumed constant	Silent, brightness-dependent flux error, worst in the faint early-time regime this project prioritises
A binary triage flag mistaken for a subtype label	Direct inspection of BTSbot metadata's column semantics before use	Training against "is this a real transient at all," not the intended 3-class question, producing plausible-looking numbers for the wrong task
Inverted entropy component	Direct comparison of mean entropy, in-distribution vs. held-out anomalous, both tiers	A published anomaly score quietly worse than it needed to be, with no failure symptom pointing at entropy specifically
Undersized CCD-edge cutouts discarding an entire alert	A live ingestion run (Section 4), not offline testing	Live alerts' light curves discarded over an unrelated cutout-shape issue

The common property across every row is that none of them would have raised an error; each would have produced a confident, plausible, wrong result. What caught them was not a single tool but three reinforcing habits: a frozen evaluation harness that could not be quietly adjusted to make a disappointing number look better (Section 3.1), golden tests written against one hand-verified real object before the reader code that would have to pass them existed, and pre-registered gates (Table 3.3) defined before the results they judge. This is the project's actual methodological contribution, and is presented as such rather than as an apology.

5.8 Limitations

Tier B is a systematically cleaner slice of the real sky than a live stream, not a representative sample of one: it is pre-filtered to brightening subtractions before a candidate ever reaches its public release (0.0% negative flux, against 3.0–3.1% measured on live, unfiltered ZTF traffic), it contains no forced-photometry row type at all, and its labels are drawn from a magnitude-limited spectroscopic follow-up programme, not from LSST's own future detection limit. Output probabilities

carry the ELAsTiCC training prior rather than the true sky prior; a live-ZTF class-distribution check run during development found a skew consistent with this (dividing out the training prior moved the predicted Ib/c share substantially toward the real sky rate), but the specific percentages from that check were computed once, live, against a database query and were never persisted to a stored artifact, so they are not restated here as verified figures (`paper/NUMBERS_GAPS.md`). LSST ingestion is built but unverified against real traffic. The image encoder was cut, not deferred: Tier B image data was never loaded across the project, and no schedule remained to build, fuse, and retrain around it once that became clear. Gate D remains unresolved, for the reason given in Table 4.3.

Chapter 6

Conclusion & Future Work

6.1 Conclusions

1. A causal, wavelength-conditioned light-curve encoder trained on simulated LSST-like data and fine-tuned on real ZTF data reaches 0.6716 macro-F1 on real, held-out test data, a 0.0454 drop from its own simulated-data ceiling of 0.7169 (Table 4.1).

2. Fine-tuning on a comparatively small real labelled set (5407 objects) recovers 0.1860 macro-F1 over a zero-shot, simulation-only baseline (Table 4.1) – the sim-to-real transfer strategy this project depends on is doing real, measurable work.

3. Host-galaxy static features improve simulated-data performance but measurably *degrade* real-data fine-tuning (Section 5); the leading explanation tested for this (missing photometric redshift) is refuted, not confirmed, and the true cause remains an open question.

4. An anomaly-detection ensemble combining reconstruction error, Mahalanobis distance, and an isolation forest reaches 0.8146 ROC-AUC on held-out simulated exotic classes and 0.7622 on real, spectroscopically confirmed non-supernova objects (Table 4.2) – a smaller but still useful score under real domain shift, not the best score on either individual set alone.

5. Classifier confidence is not a usable anomaly signal: it becomes *more* confident, not less, on inputs unlike anything the model was trained on (Table 5.1), and including it in the anomaly combination measurably hurt performance on both evaluation sets.

6. Gate D, the manual inspection of top-ranked anomalies against a published real-bogus-artifact anchor, could not be meaningfully evaluated this project: every population available was pre-cleaned of bogus candidates before the ranker ever saw it, at two independent pipeline stages (Table 4.3).

6.2 Future Work

In order of what would most directly close an open item in this report: closing Gate D against a genuinely unfiltered (or reject/candidate) alert stream; verifying the LSST reader against real traffic, whenever Fink's relay next carries it during a live session; isolating the static- encoder inversion with its own controlled ablation (a static branch trained alone, at a different learning rate, on the same Tier B split); revisiting WHXE now that Tier B fine-tuning has introduced a genuine population of non-SN-like real objects for its hierarchy to be tested against; and batching the serving state layer's per-alert database round trips, since the classifier itself is not the throughput bottleneck.

Author Contributions

A.K. led the technical work: the data pipeline and readers, model architectures and training, the evaluation harness, the anomaly-detection branch, the serving layer, and the results dashboard. A.P. led the theoretical and interpretive work: supernova progenitor physics and classification, the taxonomy adopted in Section 3.2, the astrophysical interpretation of the results in Section 5, and the literature survey in Section 2.2. Both authors contributed to writing this report.

Data and Code Availability

The pipeline's source code, configuration, tests, and this report's own figure-generation scripts live in the project repository (`sn-classifier`; repository access is internal to the project team at the time of writing). Public data sources used:

- ELAsTiCC2 training sample (LSST DESC), public, no credentials required: https://portal.nersc.gov/cfs/lsst/DESC_TD_PUBLIC/ELASTICC/ELASTICC2_TRAINING_SAMPLE_2/.
- BTSbot v10 (Zenodo), DOI 10.5281/zenodo.10839691.
- BTS Sample Explorer (Caltech), https://sites.astro.caltech.edu/ztf/bts/explorer.php.
- Fink broker, https://fink-broker.org; live ZTF and LSST alert access via `fink-client`.

References

Eric C. Bellm, Shrinivas R. Kulkarni, Matthew J. Graham, Richard Dekany, Roger M. Smith, Reed Riddle, Frank J. Masci, George Helou, Thomas A. Prince, Scott M. Adams, et al. The Zwicky Transient Facility: System overview, performance, and first results. *Publications of the Astronomical Society of the Pacific*, 131(995):018002, 2019. doi: 10.1088/1538-3873/aaecbe.

G. Cabrera-Vives, D. Moreno-Cartagena, N. Astorga, I. Reyes-Jainaga, F. Förster, P. Huijse, J. Arredondo, A. M. Muñoz Arancibia, A. Bayo, M. Catelan, P. A. Estévez, P. Sánchez-Sáez, A. Álvarez, P. Castellanos, P. Gallardo, A. Moya, and D. Rodriguez-Mancini. ATAT: Astronomical transformer for time series and tabular data. *Astronomy & Astrophysics*, 689:A289, 2024.

Tom Charnock and Adam Moss. Deep recurrent neural networks for supernovae classification. *The Astrophysical Journal Letters*, 837(2):L28, 2017. doi: 10.3847/2041-8213/aa603d.

Alexei V. Filippenko. Optical spectra of supernovae. *Annual Review of Astronomy and Astrophysics*, 35:309–355, 1997. doi: 10.1146/annurev.astro.35.1.309.

F. Förster, G. Cabrera-Vives, E. Castillo-Navarrete, P. A. Estévez, P. Sánchez-Sáez, J. Arredondo, F. E. Bauer, R. Carrasco-Davis, M. Catelan, F. Elorrieta, S. Eyheramendy, P. Huijse, G. Pignata, E. Reyes, I. Reyes, D. Rodríguez-Mancini, D. Ruz-Mieres, C. Valenzuela, L. Hernández-García, A. Jordán, A. Muñoz-Arancibia, et al. The automatic learning for the rapid classification of events (ALeRCE) alert broker. *The Astronomical Journal*, 161(5):242, 2021. doi: 10.3847/1538-3881/abe9bc.

U. C. Fremling, A. A. Miller, Y. Sharma, A. Dugas, D. A. Perley, K. Taggart, J. Sollerman, A. Goobar, M. L. Graham, J. D. Neill, J. Nordin, M. Rigault, R. Walters, I. Andreoni, A. Bagdasaryan, J. Belicki, C. Cannella, Eric C. Bellm, S. B. Cenko, K. De, R. Dekany, S. Frederick, V. Zach Golkhou, M. Graham, G. Helou, A. Y. Q. Ho, M. Kasliwal, T. Kupfer, Russ R. Laher, A. Mahabal, Frank J. Masci, R. Riddle, Ben Rusholme, S. Schulze, David L. Shupe, R. M. Smith, Lin Yan, Y. Yao, Z. Zhuang, and S. R. Kulkarni. The Zwicky Transient Facility bright transient survey I: Spectroscopic classification and the redshift completeness of local galaxy catalogs. *The Astrophysical Journal*, 895:32, 2020. doi: 10.3847/1538-4357/ab8943.

Rithwik Gupta, Daniel Muthukrishna, Nabeel Rehemtulla, and Ved Shah. Transfer learning for transient classification: from simulations to real data and ZTF to LSST. *Monthly Notices of the Royal Astronomical Society: Letters*, 542(1):L132–L138, 2025. doi: 10.1093/mnrasl/slaf074.

Željko Ivezić, Steven M. Kahn, J. Anthony Tyson, Bob Abel, Emily Acosta, Robyn Allsman, et al. LSST: from science drivers to reference design and anticipated data products. *The Astrophysical Journal*, 873:111, 2019. doi: 10.3847/1538-4357/ab042c. LSST Science Collaboration, 313 authors.

R. Kessler, G. Narayan, A. Avelino, E. Bachelet, R. Biswas, P. J. Brown, D. F. Chernoff, A. J. Connolly, M. Dai, S. Daniel, R. Di Stefano, M. R. Drout, L. Galbany, S. González-Gaitán, M. L. Graham, R. Hložek, E. E. O. Ishida, J. Guillochon, S. W. Jha, D. O. Jones, K. S. Mandel, D. Muthukrishna, A. O'Grady, C. M. Peters, J. R. Pierel, K. A. Ponder, A. Prša, S. Rodney, and V. A. Villar. Models and simulations for the photometric LSST astronomical time series classification challenge (PLAsTiCC). *Publications of the Astronomical Society of the Pacific*, 131(1003):094501, 2019. doi: 10.1088/1538-3873/ab26f1.

M. V. Kornilov, V. S. Korolev, K. L. Malanchev, A. D. Lavrukhina, E. Russeil, T. A. Semenikhin, E. Gangler, E. E. O. Ishida, M. V. Pruzhinskaya, A. A. Volnova, and S. Sreejith. Coniferest: a complete active anomaly detection framework. In *Proceedings of the Data Analytics and Management in Data Intensive Domains Conference (DAMDID 2024)*, 2024.

LSST Dark Energy Science Collaboration. ELAsTiCC: the Extended LSST Astronomical Time-series Classification Challenge. Data portal: https://portal.nersc.gov/cfs/lsst/DESC_TD_PUBLIC/ELASTICC/; code: https://github.com/LSSTDESC/elasticc, 2023. No refereed paper describing ELAsTiCC2 was found; the closest citable unit is an AAS meeting abstract (Narayan et al., 2023, AAS Meeting Abstracts, 241, 117.01).

K. L. Malanchev, M. V. Pruzhinskaya, V. S. Korolev, P. D. Aleo, M. V. Kornilov, E. E. O. Ishida, V. V. Krushinsky, F. Mondon, S. Sreejith, A. A. Volnova, A. A. Belinski, A. V. Dodin, A. M. Tatarnikov, and S. G. Zheltoukhov. Anomaly detection in the Zwicky Transient Facility DR3. *Monthly Notices of the Royal Astronomical Society*, 502(4):5147–5175, 2021. doi: 10.1093/mnras/stab316.

Anaïs Möller and Thibault de Boissière. SuperNNova: an open-source framework for Bayesian, neural network-based supernova classification. *Monthly Notices of the Royal Astronomical Society*, 2020. doi: 10.1093/mnras/stz3312. Volume/page not independently confirmed this session; cited by DOI.

Anaïs Möller, Julien Peloton, Emille E. O. Ishida, Chris Arnault, Etienne Bachelet, Tristan Blaineau, Dominique Boutigny, Abhishek Chauhan, Emmanuel Gangler, Fabio Hernandez, Julius Hrivnac, Marco Leoni, Nicolas Leroy, Marc Moniez, Sacha Pateyron, Adrien Ramparison, Damien Turpin, Réza Ansari, Tarek Allam, Jr., Armelle Bajat, Biswajit Biswas, Alexandre Boucaud, Johan Bregeon, Jean-Eric Campagne, Johann Cohen-Tanugi, Alexis Coleiro, Damien Dornic, Dominique Fouchez, Olivier Godet, Philippe Gris, Sergey Karpov, Ada Nebot Gomez-Moran, Jérémy Neveu, Stephane Plaszczynski, Volodymyr Savchenko, and Natalie Webb. Fink, a new generation of broker for the LSST community. *Monthly Notices of the Royal Astronomical Society*, 501(3):3272–3288, 2021. doi: 10.1093/mnras/staa3602.

Daniel Muthukrishna, Gautham Narayan, Kaisey S. Mandel, Rahul Biswas, and Renée Hložek. RAPID: Early classification of explosive transients using deep learning. *Publications of the Astronomical Society of the Pacific*, 131(1005):118002, 2019. doi: 10.1088/1538-3873/ab1609.

Daniel A. Perley, Christoffer Fremling, Jesper Sollerman, Adam A. Miller, Aishwarya S. Dahiwale, Yashvi Sharma, Eric C. Bellm, Rahul Biswas, Thomas G. Brink, Rachel J. Bruch, Kishalay De, Richard Dekany, Andrew J. Drake, Dmitry A. Duev, Alexei V. Filippenko, Avishay Gal-Yam, Ariel Goobar, Matthew J. Graham, Melissa L. Graham, Anna Y. Q. Ho, Ido Irani, Mansi M. Kasliwal, Young-Lo Kim, S. R. Kulkarni, Ashish Mahabal, Frank J. Masci, Shaunak Modak, James D. Neill, Jakob Nordin, Reed L. Riddle, Maayane T. Soumagnac, Nora L. Strotjohann, Steve Schulze, Kirsty Taggart, Anastasios Tzanidakis, Richard S. Walters, and Lin Yan. The Zwicky Transient Facility bright transient survey. II. a public statistical sample for exploring supernova demographics. *The Astrophysical Journal*, 904:35, 2020. doi: 10.3847/1538-4357/abbd98.

Nabeel Rehemtulla, Adam A. Miller, Théophile Jegou Du Laz, Michael W. Coughlin, Christoffer Fremling, Daniel A. Perley, Yu-Jing Qin, Jesper Sollerman, Ashish A. Mahabal, Russ R. Laher, Reed Riddle, Ben Rusholme, and Shrinivas R. Kulkarni. The Zwicky Transient Facility bright transient survey. III. BTSbot: Automated identification and follow-up of bright transients with deep learning. *The Astrophysical Journal*, 972:7, 2024. doi: 10.3847/1538-4357/ad5666.

Ved G. Shah, Alex Gagliano, Konstantin Malanchev, Gautham Narayan, Alex I. Malz, and The LSST Dark Energy Science Collaboration. ORACLE: A real-time, hierarchical, deep-learning photometric classifier for the LSST. *The Astrophysical Journal*, 995:4, 2025.

Zora Tung. ATCAT: Astronomical timeseries causal transformer. arXiv:2511.00614, 2025. No journal reference found; cited as an arXiv preprint.

V. Ashley Villar, Miles Cranmer, Edo Berger, Gabriella Contardo, Shirley Ho, Griffin Hosseinzadeh, and Joshua Yao-Yu Lin. A deep-learning approach for live anomaly detection of extragalactic transients. *The Astrophysical Journal Supplement Series*, 255(2):24, 2021. doi: 10.3847/1538-4365/ac0893.

V. Ashley Villar, Kaylee de Soto, and Alex Gagliano. Hierarchical cross-entropy loss for classification of astrophysical transients. arXiv:2312.02266, 2023. Machine Learning and the Physical Sciences Workshop, NeurIPS 2023.

Appendix A

Hyperparameters and training configuration

Table A.1: Tier A pretraining configuration (identical across the flat, LC+static, LC+static-no-photo-z, and WHXE runs except where noted).

Parameter	Value
Seed	42
Train/val/test split	0.70 / 0.15 / 0.15, stratified by class, by object id
Pre-detection window	30.0 days
Batch size	256
Epochs	15
Optimiser	AdamW, learning rate 1×10^{-3}, weight decay 1×10^{-2}
LR schedule	Cosine annealing, $T_{\max}$ = num. epochs
Class weighting	sqrt-dampened inverse frequency, normalised to mean 1
LC encoder	2-layer GRU, hidden size 128, dropout 0.2
Static encoder (LC+static only)	MLP, hidden size 32, output size 16, dropout 0.2
WHXE α (WHXE run only)	0.5
Phase cutoffs (evaluation)	$-5, 0, 5, 15, 30$, full (days since first detection)
ECE bins	15

Table A.2: Tier B fine-tuning configuration (applied identically to every fine-tuned variant: zero-shot, LC-only, LC+static, LC+static+DRB).

Parameter	Value
Seed	42
Phase 1 (epochs 1–2)	GRU trunk frozen; head (+static/DRB branch) at LR 3×10^{-4}
Phase 2 (epochs 3–8)	Trunk unfrozen at LR 3×10^{-6}; head continues at 3×10^{-4}, cosine-annealed to 0
Weight decay	1×10^{-2}
Batch size	128
Pre-detection window	30.0 days (a documented no-op on Tier B; Section 5)
Phase cutoffs (evaluation)	0, 5, 15, 30, full (Tier B has no pre-detection point)

Table A.3: Anomaly-branch reconstruction-decoder training configuration.

Parameter	Value
Seed	42
Fit subsample	60,000 Tier A objects, 3 normal classes only
Epochs	5
Batch size	256
Optimiser	AdamW, learning rate 1×10^{-3}, weight decay 1×10^{-2}
Encoder hidden size (frozen, Tier A LC-only checkpoint)	128
Decoder hidden size	64
Dropout	0.2
Mahalanobis ridge regularisation	$\varepsilon = 1 \times 10^{-3}$

Appendix B

BTS raw type-string to taxonomy mapping

Table B.1: Complete BTS Sample Explorer raw `type` string to taxonomy-id mapping. Where more than one raw string maps to the same id, they are grouped on one row.

Raw BTS type string(s)	Taxonomy id	Notes
SN Ia, SN Ia-91T	2222 (Ia)	91T-like folds into Ia; no dedicated node
SN Ia-91bg	2226 (91bg)	Dedicated node, not folded into 2222
SN Iax	2225 (Iax)	Dedicated node, not folded into 2222
SN Ib, SN Ic, SN Ic-BL, SN Ib/c	2223 (Ib/c)	
SN II, SN IIn, SN IIP, SN IIL, SN IIb	2224 (II)	IIb is II, not Ib/c
AGN	2332 (AGN)	Non-3-class, real taxonomy id
SLSN-I, SLSN-II	2242 (SLSN)	Single undifferentiated node
TDE, TDE-H-He, TDE-He, TDE-featureless	2243 (TDE)	
ILRT	2244 (ILOT)	ILRT ≈ ILOT in the literature
Ca-rich, SN Ca-rich-Ca	2245 (CART)	Moved from NO_TREE_NODE on review
Excluded from the 3-class set (`class_id=None`)		
SN Ibn, SN Icn	`excluded_csm_interacting`	Confirmed stripped-envelope, but judged CSM/interaction-powered photometrically; deliberately kept out of 2223
-, Other, other	`unclassified`	No spectroscopic classification obtained yet – missing data, not a confirmed negative
CV?, AGN?	`uncertain`	BTS's own tentative flag
SN Ia-pec, SN II-pec, SN Ib-pec, SN Ic-pec, SN IIn-pec, SN Ia-CSM, SN Ia-SC	`ambiguous`	Real spectra, family not specified unambiguously by the string alone
CV, nova, LBV, LRN	`no_tree_node`	Confirmed, unambiguous real classes with no matching taxonomy leaf